Introduction to Generative AI and Retrieval-Augmented Generation

1. What is Generative AI?
Generative Artificial Intelligence (AI) refers to systems capable of creating new content, such as text, images, audio, or video. These models learn patterns from large datasets and generate outputs that resemble human-created content. Popular examples include GPT-based models, diffusion models, and large multimodal models.

2. Types of Generative AI Models
There are several categories of generative models: • Language Models (e.g., GPT, LLaMA) • Image Generation Models (e.g., Stable Diffusion, DALL·E) • Audio Models (e.g., MusicGen) • Multimodal Models combining text, images, and video These models are trained on large datasets and use probabilistic methods to create original outputs.

3. What is Retrieval-Augmented Generation (RAG)?
Retrieval-Augmented Generation (RAG) is an AI framework that combines two powerful components: 1. A retrieval system that fetches relevant documents. 2. A generative model that uses the retrieved information to produce accurate responses. RAG helps AI overcome limitations such as hallucinations and outdated knowledge by grounding responses in real documents.

4. Why RAG is Important
RAG improves: • Accuracy • Freshness of information • Credibility • Transparency It is especially useful in academic, enterprise, legal, and scientific applications where precision is required.

5. How RAG Works
The RAG pipeline follows these steps: 1. A user asks a question. 2. The system retrieves relevant text chunks from a knowledge base. 3. The retrieved information is combined with the question. 4. The generative AI model uses both to craft a grounded response. This ensures that the answer is both relevant and evidence-based.

6. Benefits and Limitations
Benefits:
• Reduces hallucinations • Supports domain-specific tasks • Allows use of private datasets • Enhances explainability **Limitations:**
• Requires high-quality documents • Sensitive to poor chunking and embedding • Retrieval may miss important information

7. Real-World Applications
RAG is used in: • Customer support chatbots • Academic research assistants • Medical and legal information systems • Enterprise document search • Personalized tutoring systems RAG continues to evolve as organizations integrate AI into workflows requiring reliable and accurate responses.